

ENABLING INDUSTRIAL SENSORS For Edge Computing Using EdgeX

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For industrial prototyping and design, we often need industrial-grade sensors and devices plus industrial platforms like EdgeX. But interfacing and using those industrial-grade sensors with our known boards is very difficult and unknown to us. So here we use the industrial platform EdgeX and industrial sensor and interface them with the known Raspberry Pi.

Here we read the temperature and humidity values of SHT20 sensor and leverage the EdgeX framework running on Raspberry Pi to provide gateway functionality. Using EdgeX framework microservices, we process the sensor data and send it to cloud. After that we actuate LEDs to on/off when temperature readings cross threshold values. The author's prototype is shown in Fig. 1.

EdgeX framework speeds time to market by providing replaceable reference services for device-data ingestion, normalisation, edge intelligence, and sharing to support new IoT data services and advanced edge computing applications. The scope

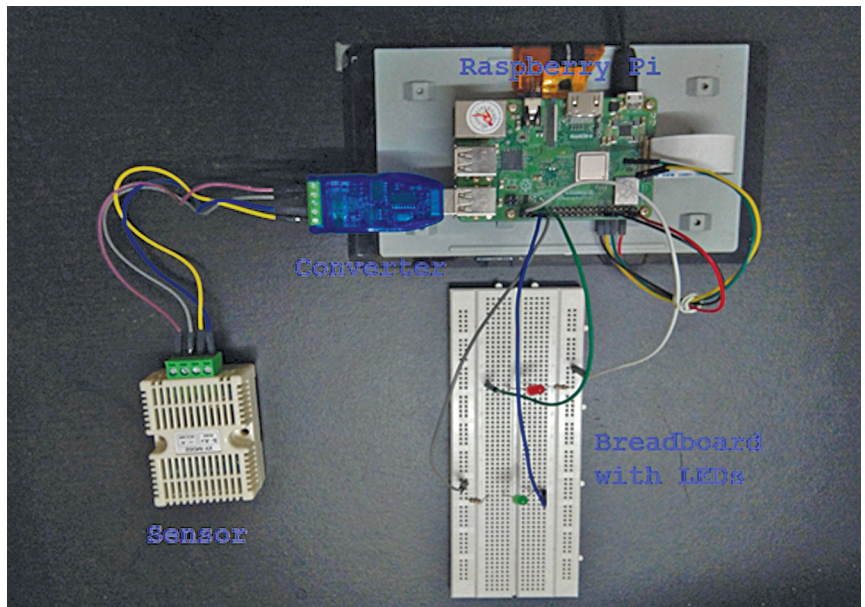


Fig. 1: Author's prototype

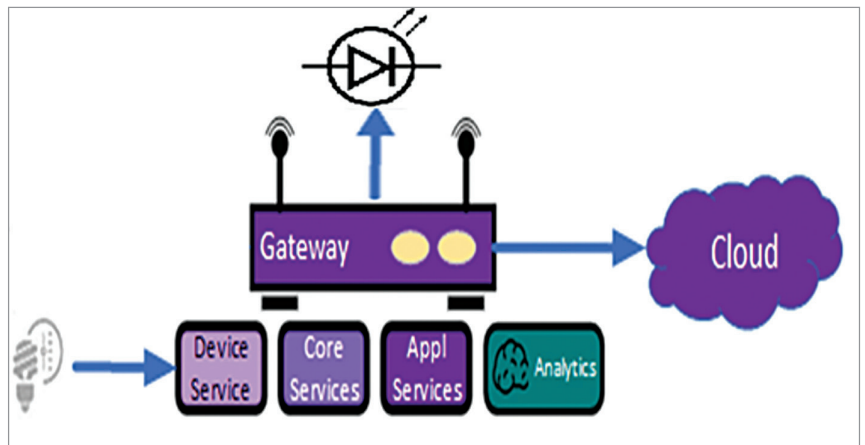


Fig. 2: High level block diagram for edge computing using EdgeX

of this case study is to leverage and exercise the functionality of EdgeX framework and demonstrate monitoring temperature and humidity sensor. Industrial grade temperature and humidity transmitter SHT20 is used for the case study.

Typical block diagram using EdgeX in Fig. 2 shows the high level block diagram for edge computing using EdgeX. It is taken from image source https://docs.edgexfoundry.org/1.3/general/EdgeX_deployments.png

BILL OF MATERIAL

Component	Quantity
Raspberry Pi 4	1
SD card	1
LED	1
Temperature and humidity transmitter Modbus SHT20 Sensor XY-MD02	1
5V, 2A DC power adaptor	1
Breadboard	1
5V DC industrial CH340 USB to RS485 converter communication module	1